

PEORIA, IL, USA

WMO: 725320

Lat: 40.668N

Lon: 89.684W

Elev: 650

StdP: 14.35

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14842

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -1.5	(c) 3.7	(d) -10.0	(e) 3.3	(f) 0.1	(g) -5.4	(h) 4.2	(i) 4.9	(j) 24.9	(k) 29.6	(l) 22.9	(m) 28.3	(n) 7.1	(o) 300	(p) 0.504

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.4	92.1	76.5	89.7	75.2	87.3	73.7	79.3	88.5	77.5	86.5	76.0	84.2	10.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.6	142.0	85.4	74.8	133.7	83.3	73.3	126.7	81.5	43.2	88.3	41.4	86.5	39.8	84.5	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.4	19.4	17.5	-8.9	96.4	6.7	2.8	-13.7	98.4	-17.7	100.1	-21.4	101.7	-26.3	103.7		
			-9.6	82.0	6.5	1.6	-14.3	83.1	-18.1	84.1	-21.8	85.0	-26.5	86.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.5	25.1	29.7	41.0	52.8	63.1	72.4	75.9	74.2	67.4	54.9	41.8	30.6
	DBStd	19.98	12.41	11.77	11.25	9.40	8.42	6.37	5.50	5.13	7.76	9.41	9.96	10.98
	HDD50	2697	774	575	323	76	5	0	0	0	1	56	281	606
	HDD65	5669	1237	989	745	379	141	14	1	3	63	336	695	1066
	CDD50	3620	1	5	45	159	412	672	803	752	523	208	36	4
	CDD65	1118	0	0	2	12	83	236	339	289	135	22	0	0
	CDH74	10099	0	0	18	139	719	2143	3319	2418	1167	174	2	0
	CDH80	3522	0	0	0	21	188	757	1292	832	400	32	0	0
Wind	WSAvg	7.9	9.0	9.0	9.3	9.8	8.2	7.1	6.1	5.6	6.2	7.6	8.6	8.3
Precipitation	PrecAvg	36.6	1.7	1.7	2.8	3.9	4.1	3.9	3.8	3.3	3.5	2.8	2.7	2.3
	PrecMax	55.4	8.1	5.3	7.5	8.2	10.4	11.7	10.2	8.6	13.1	8.0	7.6	6.3
	PrecMin	22.2	0.2	0.5	0.7	0.7	0.7	0.6	0.3	0.2	0.0	0.0	0.2	0.3
	PrecStd	8.1	1.3	1.1	1.6	1.8	2.2	2.5	2.1	2.1	2.7	1.7	1.6	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.1	65.0	76.7	83.0	88.9	93.3	97.1	93.7	92.7	84.1	71.6	62.6
		MCWB	52.5	54.9	62.6	64.8	71.7	75.1	78.3	77.0	74.1	69.1	59.9	58.3
	2%	DB	52.1	57.8	69.6	78.3	84.9	90.4	92.6	90.6	88.4	78.9	65.5	55.9
		MCWB	48.2	50.8	58.1	62.5	69.2	74.8	77.9	76.3	72.4	65.3	57.2	51.9
	5%	DB	46.2	51.4	64.0	73.4	81.6	87.6	89.8	87.6	84.7	74.5	61.3	50.4
		MCWB	42.6	46.0	55.4	59.9	67.3	73.1	76.4	74.7	70.1	63.1	55.2	46.6
	10%	DB	40.7	45.5	58.5	68.9	78.0	84.5	86.9	84.7	81.1	70.0	57.2	45.5
		MCWB	37.7	41.4	51.2	57.2	65.4	71.7	74.6	72.9	68.4	61.4	51.4	42.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.5	58.3	64.2	68.5	74.6	79.1	82.1	80.9	77.1	71.8	62.5	59.2
		MCDB	57.4	62.4	73.6	79.5	84.5	89.0	91.5	90.0	87.9	80.5	67.6	62.3
	2%	WB	48.9	51.9	60.2	64.8	72.1	77.0	79.9	78.7	74.9	68.6	59.1	52.7
		MCDB	50.9	56.9	67.6	73.6	81.6	87.1	89.1	87.5	84.5	74.9	63.3	55.0
	5%	WB	42.6	46.2	56.6	62.0	69.9	75.3	78.2	76.6	72.8	65.4	55.6	46.7
		MCDB	46.0	51.0	63.2	70.4	78.2	84.6	86.9	84.6	81.0	71.4	60.1	49.7
	10%	WB	37.8	41.3	51.2	59.0	67.7	73.5	76.6	74.9	70.6	62.5	51.8	41.9
		MCDB	40.7	45.6	57.7	68.0	75.4	82.1	84.7	82.0	78.1	69.3	56.7	45.5
Mean Daily Temperature Range	5% DB	MDBR	14.8	16.0	18.2	20.0	19.7	18.8	18.4	18.2	20.6	19.2	16.5	13.9
		MCDBR	20.1	23.0	24.0	25.8	22.8	21.5	20.3	20.7	23.5	23.9	21.8	19.1
		MCWBR	16.6	17.3	15.2	14.5	11.4	10.0	8.9	9.3	10.0	12.6	15.3	15.9
	5% WB	MCDBR	19.5	21.6	22.1	22.3	20.2	19.0	18.1	18.2	19.9	19.1	19.6	17.8
MCWBR		17.0	17.4	15.3	14.8	11.7	9.9	9.5	9.6	9.8	12.7	15.6	16.2	
Clear-Sky Solar Irradiance	taub	0.294	0.315	0.340	0.389	0.426	0.440	0.453	0.436	0.402	0.351	0.307	0.293	
	taud	2.439	2.374	2.400	2.265	2.222	2.229	2.179	2.221	2.288	2.434	2.503	2.454	
	Ebn at Noon	276	285	289	280	271	266	262	263	264	268	269	266	
	Edh at Noon	25	31	34	42	44	44	46	43	37	29	23	23	
All-Sky Solar Radiation	RadAvg	574	868	1175	1507	1742	1975	1980	1737	1478	992	660	482	
	RadStd	65	84	85	124	133	151	109	110	115	115	54	62	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+214	+133
(47) Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+175	+112

Nomenclature: See separate page